

Help & User Guide

Browse topics · search messages · inspect consumer groups and live lag · reset offsets

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Overview

Kafka Explorer is a self-hosted web UI for exploring Apache Kafka clusters. It connects to your cluster with plaintext, TLS/SSL, or SASL authentication (PLAIN, SCRAM-SHA-256, SCRAM-SHA-512, OAUTHBEARER), lists topics, searches messages by key, value, time and partition, and shows consumer group state with live lag — including the ability to reset a group's committed offsets.

- Connections support TLS with a CA certificate (ca.pem upload), mTLS client certificates, and SASL over TLS (SASL_SSL).
- Disconnect clears the saved connection from the browser and returns to the connection form.
- Message search: substring filters on key/value, time range, per-partition filter, sort by time, base64 view, JSON export.
- Consumer groups: state, members, live lag with a 5-second auto-refresh, and offset reset to earliest,

latest, timestamp, or a custom offset.

- Runs from source, in Docker, or as single-file desktop executables for Windows, macOS and Linux.

Running the app

Development

- `bun install` — install dependencies
- `bun run dev` — Vite on :5173 (proxies /api) + API on :8787
- `bun run build` — production frontend build into dist/
- `bun run server:start` — serve the built app from the API on :8787

Desktop executables

`bun run build:desktop` cross-compiled the whole app into single-file executables in `bin/` (Windows x64 .exe, macOS arm64 + x64, Linux x64). Run the file and your browser opens at `http://localhost:8787` — no Node, Bun, or Docker required. Full instructions: `EXECUTABLES.md` in the repository.

Connecting to a cluster

Open the app and pick the connection type that matches your cluster: PLAINTEXT (no security), SSL (TLS only), SASL (auth without TLS), or SASL_SSL (TLS + SASL — the common choice for managed clusters such as Confluent, Aiven, or Redpanda).

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not connected Connection settings

Optional identifier used when connecting to the cluster.

TLS / SSL

Encryption for the broker connection

CA CERTIFICATE (PEM)

-----BEGIN CERTIFICATE-----

Upload or paste your ca.pem. Needed to trust self-signed / private-CA brokers.

CLIENT CERTIFICATE (PEM) CLIENT KEY (PEM)

-----BEGIN CERTIFICATE-----

-----BEGIN PRIVATE KEY-----

Only if the broker requires client certificates (mTLS). Private key for the client certificate above.

Verify broker certificate ☒

Turn off to skip certificate validation — only for private clusters with untrusted certs

SASL authentication

Credentials the broker requires

MECHANISM

SASL/PLAIN

USERNAME PASSWORD

demo-user *****

Connection successful
Cluster demo-cluster-01 · 3 brokers · 5 topics

Test connection Save & explore topics

SASL_SSL connection form with ca.pem upload and SASL credentials

- Bootstrap brokers: one host:port per line (use the advertised listeners).
- TLS: upload your ca.pem to trust a private/self-signed CA; client certificate + key only if the broker requires mTLS.
- SASL: choose PLAIN or SCRAM-SHA-256/512 and enter username/password, or OAUTHBEARER with a bearer token.
- Test connection verifies brokers, TLS and auth before saving.
- Connection details are stored in the browser's localStorage and only sent to the local API.

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Connect to a Kafka cluster

Supports plaintext, TLS/SSL, and SASL (PLAIN, SCRAM-SHA-256/512, OAUTHBEARER). Credentials stay in your browser and are only sent to the local API.

BOOTSTRAP BROKERS

localhost:9092
kafka-1.example.com:9093

One host:port per line (or comma separated).

CONNECTION TYPE

PLAINTEXT — no encryption, no auth

Pick the security model your cluster uses (e.g. SASL_SSL for SASL over TLS).

CLIENT ID

kafka-explorer

Optional identifier used when connecting to the cluster.

Test connection Save & explore topics

Kafka Explorer · connection details are stored in your browser and used only to talk to the local API.

Connection form — choose your security model

Exploring topics

The Topics page lists every topic with its partition and replication-factor counts. Filter by name, then click a topic to open it.

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Topics

5 topics on this cluster

Filter topics... Refresh

TOPIC	PARTITIONS	REPLICATION	TYPE
orders	6	3	user
payments	4	3	user
user-events	8	2	user
inventory-updates	3	2	user
__consumer_offsets	50	3	internal

Kafka Explorer · connection details are stored in your browser and used only to talk to the local API.

Topics list

Each topic opens with two tabs: Messages (search) and Consumer groups.

Searching messages

The Messages tab filters the topic by key/value substring, time range, partition, and limit, and sorts results by time (newest or oldest first). With no filters and newest-first, only the tail of each partition is read for fast results; filtered searches scan from the beginning (capped at 250k messages). Click a row to expand the full value and headers. Use the Text/Base64 toggle for binary payloads and Export JSON to download results.

The screenshot shows the Kafka Explorer interface for the 'orders' topic. The 'Messages' tab is active, displaying search filters and results. The filters include: KEY CONTAINS (Substring of key), VALUE CONTAINS (Substring of value), FROM (LOCAL TIME) (mm/dd/yyyy, --:--), TO (LOCAL TIME) (mm/dd/yyyy, --:--), PARTITION (All partitions), LIMIT (100 messages), and SORT BY TIME (Newest first). The results table shows 12 matches, scanned 1.482,113 · 612 ms. The table columns are TIME (UTC), PARTITION, OFFSET, KEY, VALUE, and SIZE. The values are JSON objects representing order data.

TIME (UTC)	PARTITION	OFFSET	KEY	VALUE	SIZE
2026-08-11 23:13:00.730 UTC	0	4	ord-10231	{"orderId":"ord-10231","sku":"KB-2041","qty":2,"total":129.98,"status":"paid"}	87 B
2026-08-11 23:11:00.730 UTC	1	7	ord-10232	{"orderId":"ord-10232","sku":"KB-1100","qty":1,"total":59.0,"status":"paid"}	85 B
2026-08-11 23:04:00.730 UTC	2	9	ord-10233	{"orderId":"ord-10233","sku":"KB-2041","qty":5,"total":324.95,"status":"shipped"}	90 B
2026-08-11 22:59:00.730 UTC	3	3	ord-10234	{"orderId":"ord-10234","sku":"KB-9912","qty":1,"total":899.0,"status":"paid"}	86 B
2026-08-11 22:52:00.730 UTC	4	11	ord-10235	{"orderId":"ord-10235","sku":"KB-1100","qty":3,"total":177.0,"status":"cancelled","reason":"stock"}	108 B
2026-08-11 22:44:00.730 UTC	5	2	ord-10236	{"orderId":"ord-10236","sku":"KB-2041","qty":1,"total":64.99,"status":"paid"}	86 B
2026-08-11 22:37:00.730 UTC	0	5	ord-10237	{"orderId":"ord-10237","sku":"KB-8871","qty":2,"total":239.98,"status":"shipped"}	90 B
2026-08-11 22:30:00.730 UTC	1	8	ord-10238	{"orderId":"ord-10238","sku":"KB-1100","qty":1,"total":59.0,"status":"paid"}	85 B
2026-08-11 22:22:00.730 UTC	2	10	ord-10239	{"orderId":"ord-10239","sku":"KB-9912","qty":1,"total":899.0,"status":"paid"}	90 B

Message search results with filters

Consumer groups

The Consumer groups tab shows the groups subscribed to the topic: state (Stable, PreparingRebalance, Dead, ...), member count, and lag. Lag updates every 5 seconds while the tab is open — the LIVE badge shows when data was last refreshed, so you can watch consumers drain (or fall behind) in real time.

GROUP ID	STATE	MEMBERS	LAG	ACTIONS
orders-processor	Stable	3	125	Reset offsets
analytics-pipeline	Stable	4	5,221	Reset offsets
payments-settlement	Stable	2	0	Reset offsets
email-notifier	Empty	0	892	Reset offsets
inventory-sync	PreparingRebalance	2	41	Reset offsets

Consumer groups with live lag

Resetting offsets

Use Reset offsets to move a group's committed position on this topic: Earliest (start from the beginning), Latest (skip existing messages), Timestamp (first message at/after a given time), or a Custom offset (clamped to the log range). The dialog shows the previous and new offset per partition after the reset. Active consumers pick up the new position on their next fetch — consumption may duplicate or skip messages.

Reset offsets

orders-processor · orders

☒ Earliest
Reset to the first available message (log start offset).

☐ Latest
Skip all existing messages; consume only new ones.

☐ Timestamp
Reset to the first message at or after the given time.

☐ Custom offset
Reset every partition to the given offset (clamped to the log range).

This rewrites the committed offsets for this group on this topic. Active consumers will pick up the new position on their next fetch — consumption may duplicate or skip messages.

Cancel

Reset offsets

Reset offsets dialog

Security & troubleshooting

- The API is unauthenticated and binds to 0.0.0.0 — only run it on machines you trust; don't expose port 8787 to the public internet.
- Kafka credentials are stored in the browser in plain text (localStorage) — avoid production secrets in a shared browser.
- Desktop executables are not code-signed: macOS Gatekeeper (right-click !' Open) and Windows SmartScreen (More info !' Run anyway) may warn.
- Connection timeout !' broker unreachable from the machine running the app; check advertised listeners and firewall.
- Certificate error !' upload the broker CA, or disable certificate verification for private clusters.
- SASL authentication failed !' wrong mechanism or credentials; SCRAM-SHA-512 vs 256 matters.
- Port 8787 in use !' set KAFKA_EXPLORER_API_PORT.